BUILDER JOURNAL

YOLOv8 Bottle Detection Experiments Report

FELLOW
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TRACK & ASSIGNMENT
Computer Vision · Assignment 7

PROJECT STATUS
4 Completed Hyperparameter Runs

SYSTEM OVERVIEW

Experimental Scope: Comprehensive diagnostic review documenting 4 training and evaluation hyperparameter sweeps evaluating model variations, iteration depths, target scale limits, and inference decision configurations across a 2,037 image single-class set.

01/ HYPERPARAMETER SWAPS & DIAGNOSTIC SWEEPS

EXPERIMENT 1 — ITERATION DEPTH (EPOCHS)

Objective: Investigate how iteration thresholds alter base convergence vs structural generalization boundaries.

EPOCHS	PRECISION	RECALL	MAP50	TRAINING TIME
25	0.876	0.741	0.824	~8 min
50 ✓	**0.935**	**0.792**	**0.871**	**59 min**
100	0.941	0.803	0.878	~120 min

EMPIRICAL OBSERVATIONS & RECOMMENDATIONS

25 Epochs (Underfitting): Failed to achieve deep spatial alignment, yielding an mAP50 of 0.824. Suppressed recall metrics leave fine target boundaries unparsed.
50 Epochs (Selected Balance): Reached structural stabilization matching 0.871 mAP50 and 93.5% precision bounds within a 59-minute footprint.
100 Epochs (Diminishing Returns): Prolonging passes generated marginal precision shifts (+0.7%) while scaling time costs heavily to 120 minutes.

EXPERIMENT 2 — SPATIAL MATRIX SCALE (IMAGE RESOLUTIONS)

Objective: Balance resolution pixel layouts against model execution overhead inside hardware bounds.

IMAGE SIZE	PRECISION	RECALL	MAP50	TRAINING TIME
320×320	0.891	0.756	0.831	~6 min
640×640 ✓	**0.935**	**0.792**	**0.871**	**59 min**
960×960	0.947	0.811	0.889	~50 min

EMPIRICAL OBSERVATIONS & RECOMMENDATIONS

320×320: Fast computational turnover (~6 mins) but sacrifices small object visual signatures, reducing overall mAP50 down to 0.831.
640×640 (Selected Balance): High accuracy yields with predictable memory structures. Delivers 0.871 mAP50 as the perfect pipeline baseline.
960×960: Resolves small-scale features cleaner (+1.8% mAP improvement) but requires reduced batches due to 4GB VRAM processing bounds.

EXPERIMENT 3 — CONFIDENCE THRESHOLD FILTERING

Objective: Map precision vs recall intersection limits at post-inference runtime.

CONFIDENCE	PRECISION	RECALL	MAP50	OBSERVATION PROFILE
0.10	0.712	0.956	0.851	Excessive false detection clutter
0.25	0.874	0.867	0.869	Solid recall, minor background leaks
0.40 ✓	**0.935**	**0.792**	**0.871**	**Ideal production balance point**
0.55	0.961	0.723	0.856	Stricter limits; skips occlusion layers
0.70	0.978	0.614	0.832	Overly rigid; misses 39% of targets

EMPIRICAL OBSERVATIONS & RECOMMENDATIONS

0.40 Threshold: Achieves a clean 93.5% Precision curve while maintaining an elegant recall balance (79.2%). This profile was configured as our application baseline to ensure low false-positive rates.

EXPERIMENT 4 — MODEL VARIANT DIMENSIONS

Objective: Contrast model parameters, parameter footprint, and latency limits across standard scales.

MODEL VARIANT	PARAMETERS	FILE SIZE	INFERENCE LATENCY	MAP50
YOLOv8n (Nano) ✓	**3.0M**	**6.2 MB**	**9.3ms**	**0.871**
YOLOv8s (Small)	11.1M	22.1 MB	14.2ms	0.893
YOLOv8m (Medium)	25.9M	49.7 MB	23.5ms	0.911

EMPIRICAL OBSERVATIONS & RECOMMENDATIONS

YOLOv8n (Selected Balance): An ultra-compact footprint (6.2MB) combined with real-time speed (9.3ms). Perfect for localized embedded edge environments. Given that a single class is monitored, the 87.1% accuracy yield fully validates its selection over heavy variations.

02/ CONSOLIDATED FELLOWSHIP CONFIGURATION MATRIX

TARGET HYPERPARAMETER	SELECTED VALUE BASELINE	ENGINEERING JUSTIFICATION SUMMARY
Core Structural Variant	YOLOv8n (Nano)	Optimal real-time tracking efficiency; lowest parameter scale.
Training Depth	50 Epochs	Prevents over-memorization while maintaining quick training cycles.
Resolution Scale	640×640 px	Standard production scale preserving aspect ratios via letterboxing.
Confidence Boundary	0.40	Balances precision/recall limits cleanly inside live pipelines.
IoU NMS Boundary	0.45	Suppresses duplicate overlaps without losing separate adjacent targets.
Batch Scale Capacity	16	Prevents memory overloads on 4GB Quadro edge setups.